

Lecture 4

09/08/24 9:32 AM

BFS - bipartiteness testing
shortest paths (unweighted edges)
DFS - checking for cycles
topological sort

SHORTEST PATH (weighted edges)

Consider a graph $G = (V, E)$.

We want to find the shortest path b/w vertices s and t .

If all edge weights are equal, then BFS gives us shortest paths.

SHORTEST PATHS IN DAGs: → directed acyclic graphs

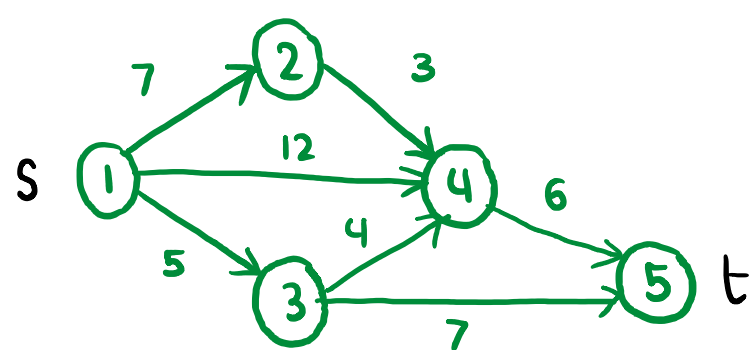
$G = (V, E)$

weight function: $wt: E \rightarrow \mathbb{R}$

If P is a path with edges $e_1, e_2, \dots, e_k$ in it, then

$$wt(P) = wt(e_1) + wt(e_2) + \dots + wt(e_k).$$

Q. Find the shortest distance from 1 $\rightsquigarrow$ 5.



Place the nodes in topological order.

• ALGORITHM:

$\forall v \in V \setminus \{s\}, dist(s, v) \leftarrow \infty$

Visited $\leftarrow \{s\}$

While Visited $\neq V$: → neighbors (may be visited already or not)

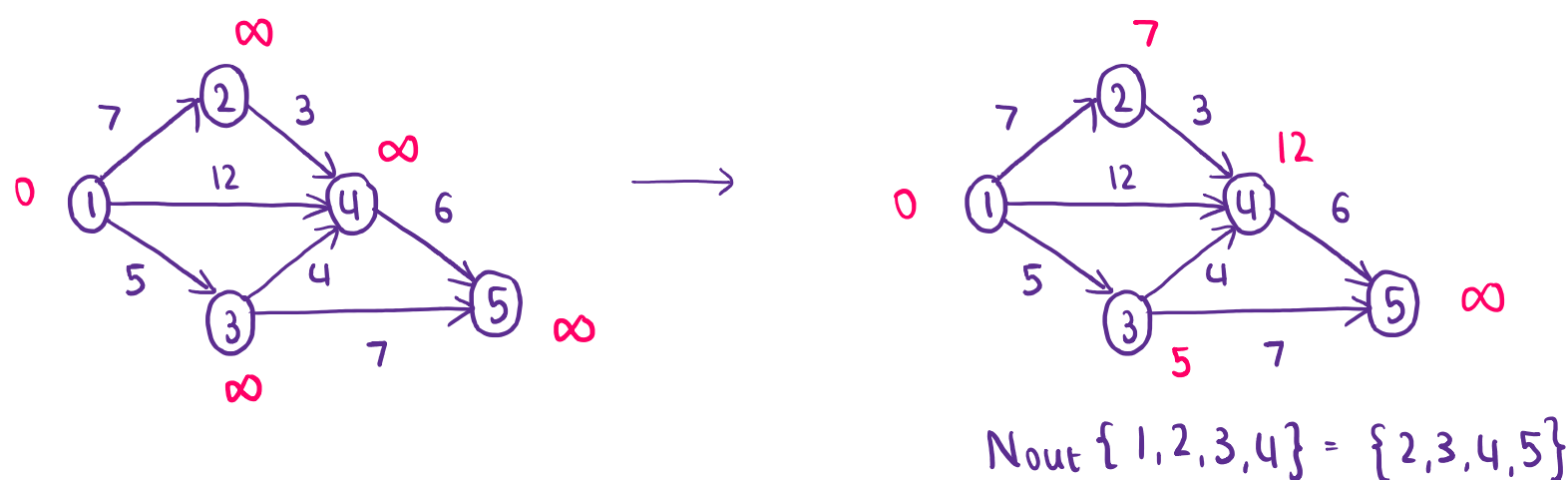
For all $v \in N^{out}(Visited)$: → eventually terminates since DAG

$$dist(s, v) = \min \left\{ \begin{array}{l} \{ dist(s, u) + wt(u, v) \mid \substack{(u, v) \in E \\ u \in Visited} \} \\ \cup \{ dist(s, v) \} \end{array} \right\}$$

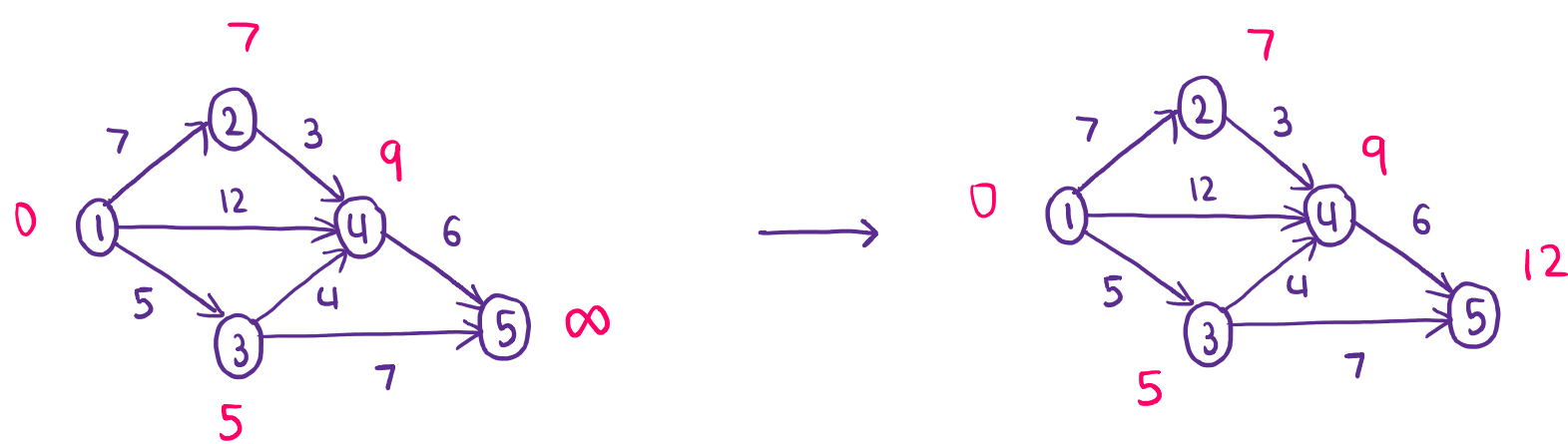
Visited $\leftarrow Visited \cup \{v\}$

not
precise

- multiple vertices may be visited in the same iteration.
- distances may change in later iterations.



$$dist(1, 4) = \min \left\{ \begin{array}{l} 12 \\ dist(1, 2) + wt(2, 4) \\ dist(1, 3) + wt(3, 4) \end{array} \right\} = 9$$



$$dist(1, 5) = \min \left\{ \begin{array}{l} 15 \\ dist(1, 4) + wt(4, 5) \\ dist(1, 3) + wt(3, 5) \end{array} \right\} = 12$$

FAILS in the case of
cyclic dependencies

SINGLE SOURCE GRAPHS with no negative edges:

• DIJKSTRA'S ALGORITHM

works for undirected graphs also

Visited $\leftarrow \{s\}$

For every $v \in V \setminus \{s\}$

$d(v) \leftarrow \infty$

While visited $\neq V$:

For all $v \in \boxed{(V \setminus Visited) \cap N(Visited)}$: A

$$d'[v] = \min_{(u, v) \in E} \{ d(u) + wt(u, v) \}$$

Select $v \in A$ s.t. $d'[v] = \min \{ d'[v] \mid v \in A \}$

Set $d[v] \leftarrow d'[v]$.

Add v to Visited.

- only one vertex visited per iteration
- calculated distances do not change in later iterations.

• OBSERVATIONS:

1. Visited set grows but elements once added are not disturbed in the later stage of the algorithm.
2. Shortest distance once computed are not updated ever again.

$$O((V+E) \log V)$$